Test 3 review day

CMSE 381

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Announcements

Announcements:

- midterm # 3 on Monday
- Bonus code portfolio assignment due Nov 26 on D2L.
- Final Project submission on D2L. Due Dec 5.

Based on the questions submitted

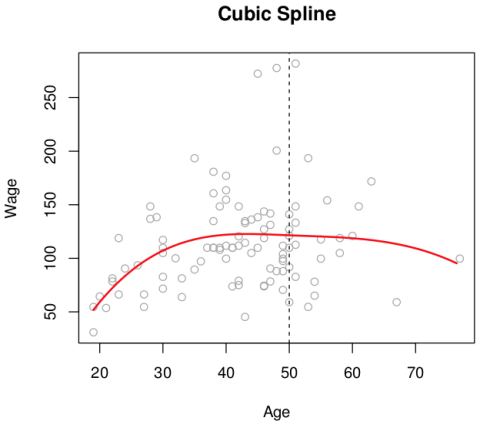
- Cubic spline math
- Bagging and random forests
- SVM kernels
- Convolutional and softmax output

Section 1

Cubic Splines

Cubic splines

- Fit piecewise polynomial
- Require continuity at knots
- Require the first and second derivatives to be continuous at knots



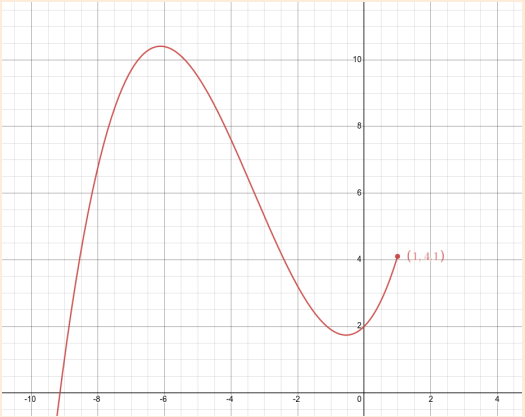
Test your understanding: PollEv

Example

We have the following piecewise cubic polynomial:

$$f(x) = \begin{cases} 2 + x + x^2 + 0.1x^3 & x \leq 1 \\ b_0 + b_1 x + b_2 x^2 - x^3 & x > 1 \end{cases}$$

What are b_1, b_1, and b_2 to make this a cubic spline?

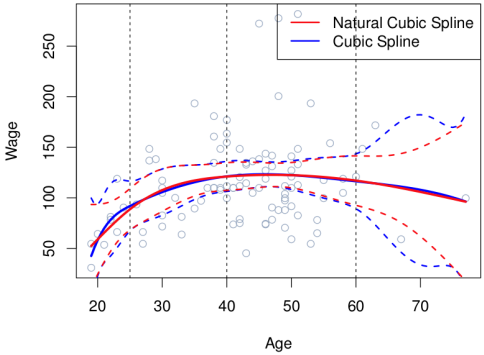


Check your answers: www.desmos.com/calculator/kbm0zivqco

More space for work

$$f(x) = \begin{cases} 2 + x + x^2 + 0.1x^3 & x \leq 1 \\ b_0 + b_1 x + b_2 x^2 - x^3 & x > 1 \end{cases}$$

Notes on cubic splines



Section 2

Bagging and Random Forests

Use ensemble of trees to reduce variance

Want to do (but can't):
Build separate models from independent training sets, and average resulting predictions:

- $\hat{f}^1(x), \cdots, \hat{f}^B(x)$ for B separate training sets
- Return the average

$$\hat{f}_{avg}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^b(x)$$

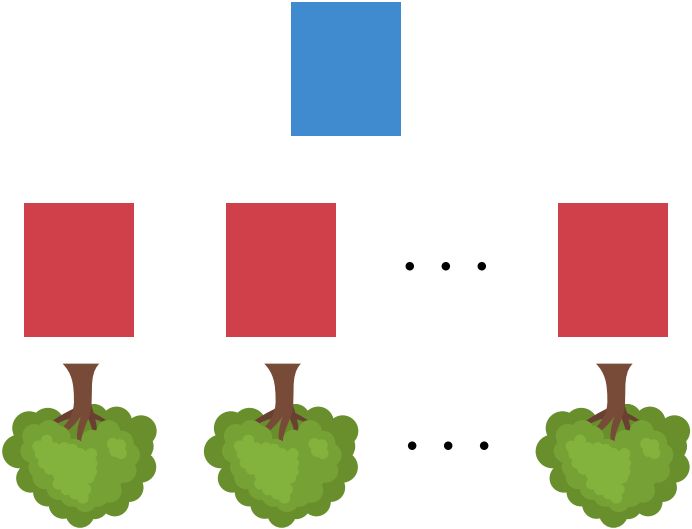
Boostrap modification:

- Work with fixed data set
- Take B samples from this data set (with replacement)
- Train method on bth sample to get $\hat{f}^{*b}(x)$
- Return average of predictions (regression)

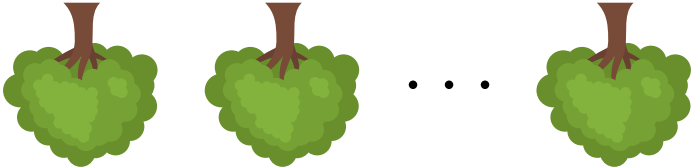
$$\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{*b}(x)$$

or majority vote (classification)

Tree version

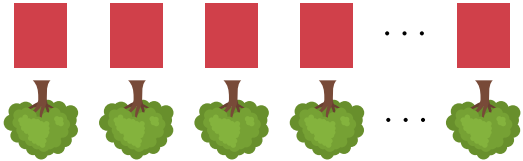


Prediction on new data point



Out of Bag Error Estimation

- On average, bootstrap sample uses about 2/3 of the data
- Remaining observations not used are called *out-of-bag* (OOB) observations
- For each observation, run through all the trees where it wasn't used for building
- Return the average (or majority vote) of those as test prediction! rather law of large number
- Bootstraped version of LOOCV.



Random Forests

- Goal is to decorrelate the bagged trees:
  - If there is a strong predictor, the first split of most trees will be the same
  - Most or all trees will be highly correlated
  - Averaging highly correlated quantities doesn't decrease variance as much as uncorrelated

- The random forest fix:
  - Each time a split is considered, only use a random subset of m the predictors
  - Fresh sample taken every time
  - Typically $m \approx \sqrt{p}$
  - On average, $(p - m)/p$ of splits won't consider strong predictor
  - $m = p$ gives back bagging

Section 3

SVM Kernels

The kernel - replacing global coordinate with local similarity

$$y_i f(x_i) \geq M(1 - \epsilon_i), \text{ where}$$

$$K(x_i, x_i')$$

$$f(x) = \beta_0 + \sum_{j \in \mathcal{S}} \alpha_i K(x, x_j)$$

SVC via inner products

$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle x, x_i \rangle$$

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \langle x, x_i \rangle$$

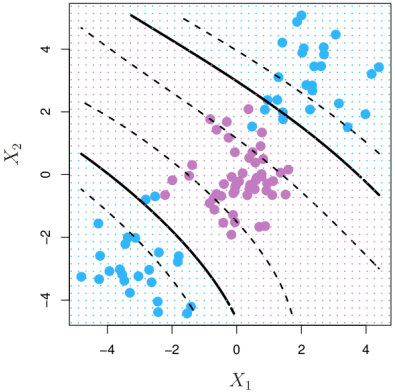
The kernel - more generally

$$K(x_i, x_i')$$

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

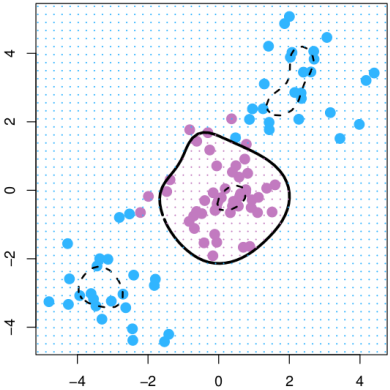
A polynomial kernel

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$



A radial kernel

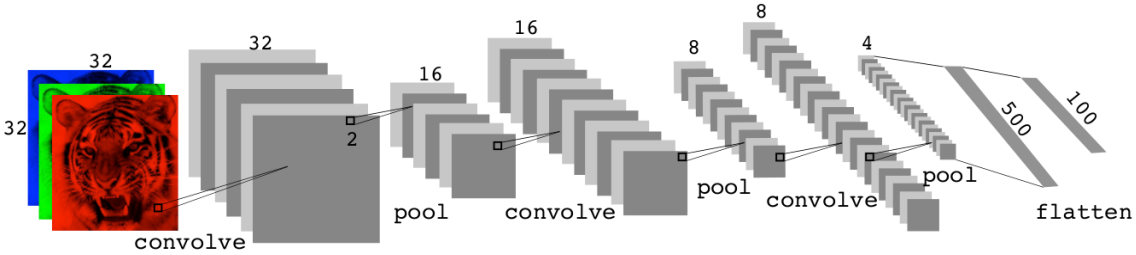
$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$



Section 4

Convolutional Neural Networks and softmax output

CNN



https://poloclub.github.io/cnn-explainer/

Convolution layer

Convolution Filter

Original Image:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{bmatrix}$$

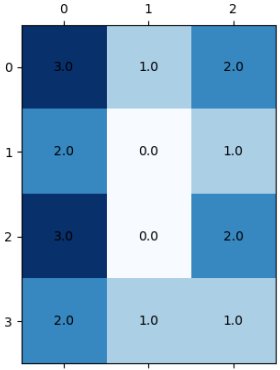
Convolution filter:

$$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

Convolved Image

$$\begin{bmatrix} a\alpha + b\beta + d\gamma + e\delta & b\alpha + c\beta + e\gamma + f\delta \\ d\alpha + e\beta + g\gamma + h\delta & e\alpha + f\beta + h\gamma + i\delta \\ g\alpha + h\beta + j\gamma + k\delta & h\alpha + i\beta + k\gamma + l\delta \end{bmatrix}$$

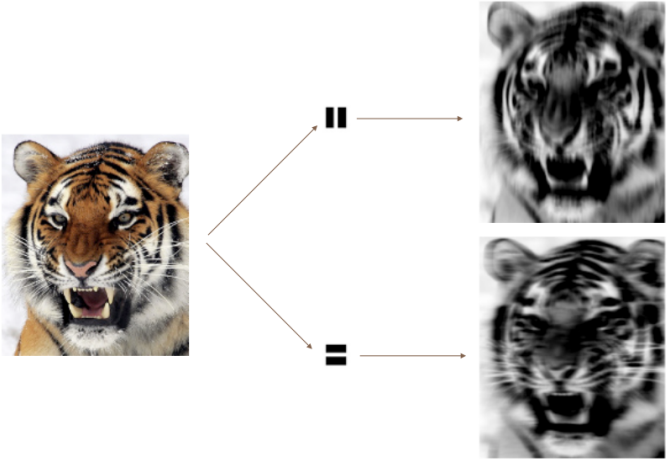
Convolution Filter Example



Filter:

$$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

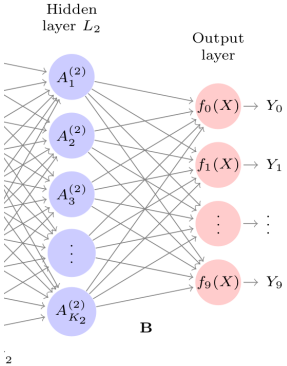
Convolution filter: Bigger example



Pooling layers

$$\text{Max pool} \begin{bmatrix} 1 & 2 & 5 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$$

The last column for classification: Softmax



$$f_m(X) = \Pr(Y = m|X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},$$